

Decision Trees and Rule-Based Learning

This concise module is designed for AI-agent based learning and quick revision.

Decision Tree

A supervised learning algorithm that splits data into decision nodes to make predictions.

Entropy

Measures impurity of a dataset. Lower entropy means purer data.

Information Gain

Chooses the attribute that best splits the data.

Gini Index

Alternative impurity measure used by CART.

ID3

Builds trees using Information Gain.

C4.5

Improves ID3 by handling continuous values and pruning.

CART

Uses Gini Index and supports classification and regression.

Random Forest

An ensemble of many decision trees for improved accuracy.

Quick Revision: Revise definitions first, understand key concepts, then practice MCQs.